

Electronic Balancing Circuit for a Crossed Coil NMR Spectrometer

Ronald H. Geils

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is fed back to the base of Q2, cuts off Q2. Thus, all the current I flows through Q1. Since no current flows through Q2, there is no voltage drop across resistor R3 and the full voltage appears at output 1, as indicated in Fig. 2.

When the drive tunnel diode is at its high voltage state (approximately 0.5 V), Q1 is cut off. The resultant zero voltage drop across R2 causes Q2 to conduct all the current I . This current, flowing through R3, yields the negative voltage at output 1, as indicated in Fig. 2.

It will be noted that when Q1 conducts, its base current (I_{b1}) flows through the driving tunnel diode. To assure correct operation of the driving tunnel diode, this Q1 base current must be kept small. Thus,

$$\begin{aligned} I_{b1} &= I_m, \text{ or} \\ I &= I_m \cdot B, \end{aligned} \quad (1)$$

where I_m is the maximum value of I_{b1} which has negligible effect on the operation of the driving tunnel diode and B is the minimum current gain of Q1.

With I determined from Eq. (1), and since the voltage at the emitters of Q1 and Q2 clamps at approximately 0.7 V,

$$R1 = (| -V | - 0.7) / I. \quad (2)$$

Also, since the voltage drop across R2 (when Q1 conducts) should be approximately 0.25 V (one-half the voltage across the drive tunnel diode in its high voltage state),

$$R2 = 0.25 / I. \quad (3)$$

To assure that the voltage at output 1 swings from $+V$ to approximately -0.5 V,

$$R3 = (| +V | + 0.5) / I. \quad (4)$$

The component values shown in the circuit of Fig. 1 were determined using Eqs. (1)–(4), with the assumptions that a negligible value of Q1 base current $I_m = 0.1$ mA, and the minimum current gain B of Q1 = 30. The supply voltages used were $-V = -2.0$ V and $+V = +2.0$ V.

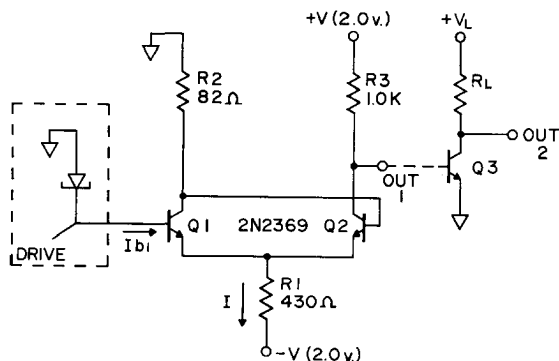


FIG. 1. Tunnel diode output voltage converter, schematic diagram.

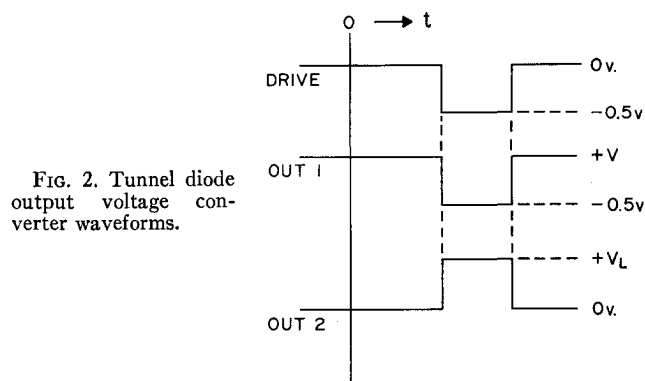


FIG. 2. Tunnel diode output voltage converter waveforms.

The full voltage swing at output 1, indicated by Fig. 2, will be obtained only if the load added at output 1 draws a small fraction of the current I . Where larger currents through the output load are required, a buffer driving stage, such as that shown in Fig. 1 with the output waveform indicated in Fig. 2, may be added.

¹ C. A. Cancro and N. M. Garrahan, *Rev. Sci. Instrum.* **40**, 1360 (1969).

² C. A. Cancro and N. M. Garrahan, *Proc. IEEE* **56**, 1389 (1968).

Electronic Balancing Circuit for a Crossed Coil NMR Spectrometer*

RONALD H. GEILS

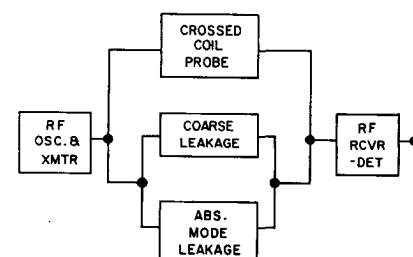
*Department of Metallurgy and Mining Engineering and
Materials Research Laboratory, University of Illinois at Urbana-
Champaign, Urbana, Illinois 61801*

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OF the common NMR detection techniques the crossed coil¹ offers the best combination of stability, optimum signal-to-noise ratio, and freedom in the selection of the level of the rf field H_1 . In this method one circularly polarized component H_c of H_1 induces a nuclear magnetization² $M_n = \chi(\omega)H_c$, where the nuclear susceptibility $\chi(\omega) = \chi'(\omega) - i\chi''(\omega)$. M_n in turn induces voltages into the receiver coil proportional to χ' and χ'' . The phases of these voltages, relative to the transmitter voltages, are found to be ± 180 and $+90^\circ$, respectively.

Although the flux-paddle method (used by Varian Associates) of balancing the probe and providing leakage is widely used, it is not suitable for low temperature work nor is it within the technical means of simply equipped laboratories to produce. The electronic balancing system

FIG. 1. Block diagram of magnetic resonance detection circuit with leakage path included.



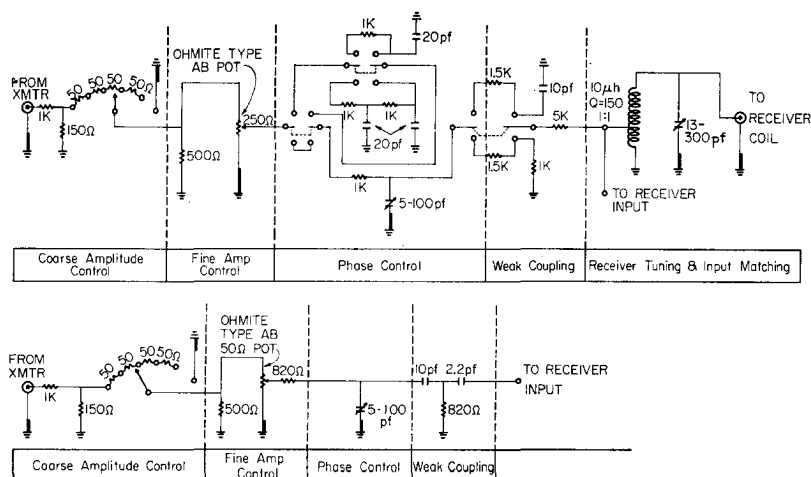


FIG. 2. Leakage circuits. Upper: coarse leakage, phase of output voltage variable from 0 to -180° relative to transmitter output; lower: absorption mode leakage, output phase near $+90^\circ$, variable capacitor adjusted as described in text. $C_{\max}$ corresponds to about 4 MHz, $C_{\min}$ to 12 MHz. The heavy lines denote very low rf impedance paths.

of Drain³ has been very successful, but requires two more coils to be wound near the receiver coil to produce the leakage. The *v*-mode electronic balancing system of Blume⁴ has both simplicity and stability, and it is this technique that we extend.

The circuits to produce the appropriate leakage voltages consist simply of passive networks coupling the transmitter output voltage to the receiver coil after suitable attenuation and phase control. Passive networks were used to achieve optimum stability, although their use limited the leakage voltages available to about 10^{-3} of the transmitter voltage. Thus the crossed coil probe used must be geometrically capable of an attenuation of about 10^{-3} , so that the residual imbalance can be eliminated electronically. This is important because in low temperature work, for example, cooling the probe changes the geometrical coupling. It has been found that crossed coil probes with the above requirement can be built without undue difficulty, provided a Faraday shield is used between the coils.

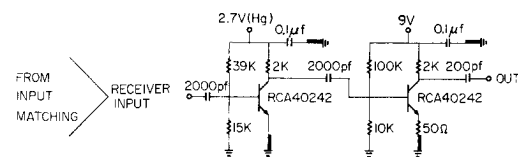
A block diagram of the circuit is shown in Fig. 1. The detailed circuits are shown in Figs. 2 and 3. The two leakage circuits were built in the same rf box as the receiver to avoid the use of connecting cable. The dashed lines indicate a grounded metallic partition in the rf box. All grounds indicated are made to the internal ground plane of the box; those indicated by heavy lines must have very low rf impedance. Each leakage circuit consists of an amplitude control, a phase control, and a weak coupling circuit. For a given frequency, the phase of the voltage from the absorption circuit is set near $+90^\circ$ from the transmitter voltage (but not exactly, as we shall see). The phase of voltage from the coarse leakage circuit is variable from about 0 to -180° from the transmitter voltage; thus any small leakage into the receiver owing to the inductive and capacitive coupling between the two coils can be nulled. Then by changing the absorption mode amplitude control, one introduces a voltage in phase

with the nuclear absorption signal (X'') permitting it to be detected.

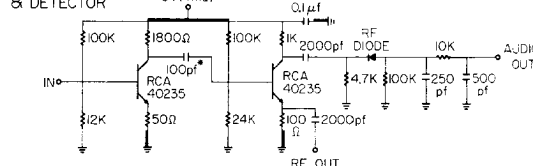
There is one further complication which is introduced because of the need for cables leading to the transmitter coil and from the receiver coil to the receiver. Let us consider a case at 10 MHz. Typically the electronic balancing signal need only travel about 30 cm from the power amplifier output to the receiver input where the mixing takes place. If cables about 1.5 m long are used, the total delay due to the cables is about 10^{-8} sec, and the NMR absorption signal will be about 60° from the power amplifier voltage at the input to the transmitter cable instead of the 90° it would be for zero cable length.

The precise phase setting for the absorption mode leakage is best determined empirically as a function of

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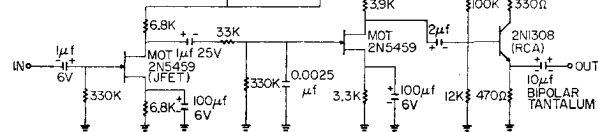


FIG. 3. Receiver circuit. The transistor biases may need adjustment owing to variations in characteristics. The 100 pF capacitor (*) may be reduced to improve the flatness of the rf amplifier from about 2 to 20 MHz. A diode with low forward resistance (such as Fairchild 6666 or 1N99A) keeps the stage gain near unity and helps in detecting an rf null. Heavy lines indicate low rf impedance.

frequency. This can be done by displaying on an oscilloscope the resonance of a material known to have a symmetrical absorption line. The absorption phase control is then set to give a symmetrical line after unbalancing. This position can be marked and the process repeated over the frequency range of interest. There is a slight dependence of the phase of the absorption leakage on the Q of the receiver coil. However, the use of weak capacitive coupling brings the phase shift of this last stage near $+90^\circ$ so that the effect is small; placing the leakage on the low side of the input matching transformer further reduces this problem. The values of the components given in Fig. 2 cover the range from 4 to 10 MHz. Suitable adjustment of the fixed values in the final stage easily varies this range.

The receiver used was completely transistorized and battery powered to gain the stability of these devices. The circuit is shown in Fig. 3. The input impedance of the receiver is about $500\ \Omega$; it was kept low so that the phase of the absorption leakage circuit did not vary strongly with the Q of the receiver coil when metallic samples were inserted or when the probe was cooled. The input equivalent noise resistance of the 40242 transistor is about $100\ \Omega$, so that the noise figure of the receiver does not increase appreciably. The over-all spectrometer is stable and reliable while also being easy to use and inexpensive to build.

The author wishes to thank Professor T. J. Rowland for his suggestion of the above technique and the many discussions about these matters.

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¹ F. Block, W. W. Hansen, and M. Packard, *Phys. Rev.* **70**, 474 (1946); E. R. Andrew, *Nuclear Magnetic Resonance* (Cambridge U. P., Cambridge, England, 1955).

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³ L. E. Drain, *Discussions Faraday Soc.* **19**, 200 (1955).

⁴ R. J. Blume, *Rev. Sci. Instrum.* **33**, 1472 (1962).

Magnetic Field Measurements Using an Electron Beam and an Electrostatic Analyzer*

JOHN S. RISLEY†

Department of Physics, University of Washington,
Seattle, Washington 98105

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THIS note describes a procedure by which an electron beam in an electrostatic analyzer is used as a magnetometer. The measured apparent energy of the electrons was affected by changes in the current to a pair of Helmholtz coils controlling the component of the magnetic field perpendicular to the electron beam trajectory. A method is set forth in this note to determine the correct current to

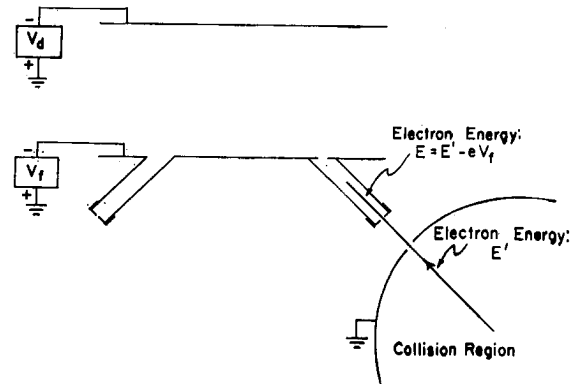


FIG. 1. The experimental configuration used to measure the effect of the magnetic field inside an analyzer.

the Helmholtz coils in order to cancel this component of the magnetic field. An approximate expression for the effect of a weak magnetic field (less than $\frac{1}{2}$ G for 2 eV electrons) on the operation of a parallel plate analyzer¹ was derived and used to exhibit the first order dependence of the magnetic field on the calibration constant C (C =deflection voltage/kinetic energy) for a cylindrical mirror analyzer.²⁻⁶

Electrons which passed through a field-free region with energy E' were accelerated or decelerated by a voltage V_f on the front plate of the analyzer (see Fig. 1), obtaining a kinetic energy $E = E' - eV_f$ at the front surface of the analyzer.⁷ Inside the analyzer the total voltage deflecting the electrons was $V_d - V_f$ where V_d is the voltage on the back plate; thus $e(V_d - V_f) = CE$ and

$$V_d = \frac{E'}{e} + \frac{C-1}{C} V_{an}, \quad \text{where } V_{an} = V_d - V_f. \quad (1)$$

In practice, a voltage supply was connected between the front and the back plates which allowed the analyzing voltage V_{an} to be varied and measured directly. For zero magnetic field, C is simply a geometrical proportionality constant; however, in the presence of a magnetic field B , an expression was derived for a parallel plate analyzer which showed that C should be replaced by $C^* \approx [1 + kB/(eV_{an})^{\frac{1}{2}}]C$ where k is another geometrical constant. If $kB/(eV_{an})^{\frac{1}{2}} \ll 1$, Eq. (1) becomes, approximately,

$$eV_d \approx E' + [(C-1)/C]eV_{an} + (k/C)(eV_{an})^{\frac{1}{2}}B. \quad (2)$$

The magnetic field B normal to the path of the electrons inside the analyzer will cause the electrons to either overshoot or undershoot the distance to the exit hole. By readjusting the deflection voltage, the electrons can be redirected to pass through the exit hole. The field was varied by adjusting the current to the external Helmholtz coils. Hall probes were used to calibrate the differential magnetic field changes produced by the coils. The change in the deflection voltage at which these monoenergetic